

Named Entity Recognition Using BERT for Automatic Entity Extraction

Abstract

Named Entity Recognition (NER) is a fundamental Natural Language Processing (NLP) task that aims to identify and classify named entities such as persons, organizations, and locations within text. This project develops an NER system using a fine-tuned BERT (Bidirectional Encoder Representations from Transformers) model trained on the WikiAnn dataset. The system employs tokenization, BIO tagging, label alignment, and transfer learning techniques to accurately recognize entities in English text. Experimental results demonstrate that the proposed model achieves an F1-score of 82.75%, precision of 81.78%, recall of 83.75%, and token-level accuracy of 92.06% on the test dataset. The trained model is further deployed through a Gradio web application for interactive entity extraction.

Keywords: Natural Language Processing, Named Entity Recognition, BERT, Token Classification, Deep Learning, Transformers

I. Introduction

Named Entity Recognition (NER) is one of the most important tasks in Natural Language Processing (NLP). It focuses on identifying and classifying entities such as person names, organizations, locations, dates, and other meaningful information within unstructured text.

For example, in the sentence:

"Elon Musk founded SpaceX in California."

an NER system should recognize:

- Elon Musk → Person (PER)
- SpaceX → Organization (ORG)
- California → Location (LOC)

NER plays a critical role in various real-world applications including information extraction, question answering systems, search engines, recommendation systems, chatbots, and document analysis.

Traditional NER approaches relied heavily on handcrafted features and statistical models. However, recent advances in deep learning and transformer architectures have significantly improved NER performance. In particular, BERT has become one of the most successful pre-trained language models for sequence labeling tasks.

The objective of this project is to build and evaluate a BERT-based NER system capable of automatically identifying named entities from English text.

II. Dataset Description

The project utilizes the WikiAnn dataset provided through the Hugging Face Datasets library.

Dataset:

- WikiAnn (English)
- Source: Hugging Face Datasets
- Language: English

Each sample contains:

Field	Description
tokens	List of words in a sentence
ner_tags	Entity labels in BIO format
langs	Language code
spans	Entity span information

Example:

Tokens:

["Barack", "Obama", "visited", "France"]

NER Tags:

["B-PER", "I-PER", "O", "B-LOC"]

The dataset follows the BIO tagging scheme:

- B-PER = Beginning of Person entity
- I-PER = Inside Person entity
- B-ORG = Beginning of Organization entity
- I-ORG = Inside Organization entity
- B-LOC = Beginning of Location entity
- I-LOC = Inside Location entity
- O = Outside any entity

III. Methodology

A. Data Preparation

The dataset was loaded using the Hugging Face Datasets library:

```
from datasets import load_dataset
dataset = load_dataset("unimelb-nlp/wikiann",
"en")
```

The label names were extracted from the dataset metadata to map integer labels into human-readable BIO tags.

B. Tokenization

The pre-trained BERT tokenizer was employed to convert words into subword tokens.

Model: ***bert-base-cased***

Example:

Original word: Washington

Tokenized output:

Wash
##ington

Tokenization is necessary because BERT processes subword units rather than complete words.

C. Label Alignment

Label alignment is one of the most important stages in BERT-based NER systems.

Since a single word may be split into multiple subword tokens, each generated token must receive a corresponding label.

Example:

Original:

Washington → B-LOC

Tokenized:

Wash
##ington

Aligned labels:

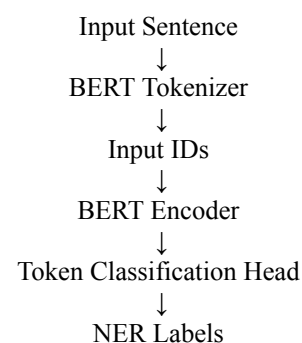
Wash → B-LOC
##ington → I-LOC

Special tokens such as [CLS] and [SEP] were assigned the value -100 to ensure they are ignored during loss computation.

D. Model Architecture

The project employs BERT for token classification.

Architecture:



The model was initialized using:

AutoModelForTokenClassification

with seven output labels corresponding to the BIO tagging scheme.

E. Training Configuration

The model was trained using a Hugging Face Trainer.

Training parameters:

Parameter	Value
Learning Rate	3e-5
Batch Size	16
Epochs	10
Weight Decay	0.01
Optimizer	AdamW
Evaluation Strategy	Every Epoch

The best model was automatically selected according to the highest validation F1-score.

F. Evaluation Metrics

The seqeval evaluation library was used to compute NER-specific metrics. The following metrics were measured:

Precision: Measures how many predicted entities are correct.

Recall: Measures how many actual entities are successfully identified.

F1-Score: Harmonic means of precision and recall.

Accuracy: Measures token-level prediction correctness.

IV. Experimental Results

The model was evaluated on the test dataset.

Results:

Metric	Score
Evaluation Loss	0.2601
Precision	81.78%
Recall	83.75%
F1-Score	82.75%
Accuracy	92.06%

Discussion

The experimental results indicate that the BERT-based model successfully learns contextual representations and accurately identifies named entities.

The high recall score suggests that the model effectively captures most entities present in the text, while the precision score indicates relatively few false-positive predictions.

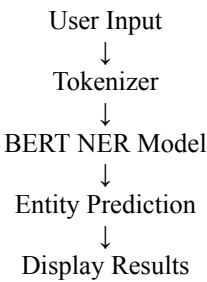
An F1-score of 82.75% demonstrates strong overall performance and confirms the effectiveness of transfer learning through pre-trained transformer models.

The evaluation loss of 0.26 further indicates that the model converged successfully during training and generalizes well to unseen examples.

V. Deployment

To make the model accessible to users, a Gradio-based web application was developed.

Workflow:



Users can enter any English sentence and receive real-time entity recognition results through a simple web interface.

VI. Conclusion

This project successfully implemented a Named Entity Recognition system using a fine-tuned BERT model. Through tokenization, label alignment, BIO tagging, and transformer-based token classification, the model achieved strong performance on the WikiAnn dataset.

The experimental results demonstrated an F1-score of 82.75% and an accuracy of 92.06%, indicating the model's capability to effectively recognize persons, organizations, and locations from natural language text.

Future work may include experimenting with larger transformer architectures, multilingual NER datasets, CRF-based decoding layers, and domain-specific datasets such as medical or financial text.

References

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